The Khanna–Loehr local identity for t-weighted rim hooks: a counterexample, a corrected classification, and a certificate family

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Abstract

Let $\mathrm{wt}_A(\lambda, \gamma) = t^{\mathrm{ht}(\lambda/\gamma)}$ be the t-weight on rim-hook removals and, for $\mu \vdash n$, consider the Khanna–Loehr local identity with the symmetric choice $T(\mu, L) = S(\mu, L)$ and completely free weights $\mathrm{wt}_B(\mu, \gamma) \in \mathbb{Q}(t)$. My paper of this morning, `2026-09-10-Q129-reciprocity-does-not-deform.tex`, asserts (Theorem `thm:local`) that this system is inconsistent over $\mathbb{Q}(t)$ for $4 \le n \le 7$ and *every* $\mu \vdash n$, and records that as verified but not proved.

The theorem is **false**. At $n = 4$, $\mu = (2, 2)$ the system has the explicit solution $\big(-1,\ t,\ 1,\ -t\big)/(t-1)^2$; at $n = 5$ it is solvable for $\mu = (3, 2)$ and $\mu = (2, 2, 1)$. All three are hand-checkable and are contradicted by the anchor-scan table printed in that same paper, which I reproduce here on the nose with an independently built instrument.

I then prove what is true. Writing $C(\mu)$ for the set of rim-hook predecessors of μ:

- $|C(\mu)| = n$ exactly, so the system is $p(n)$ equations in n unknowns (Proposition 2.1) — never square for $n \ge 4$, correcting the explanation offered for the exceptions in the earlier paper.
- At $t = -1$ the system is solvable for every μ and every n, with the closed-form solution $\mathrm{wt}_B(\mu, \gamma) = (-1)^{\mathrm{ht}(\mu/\gamma)}/n$; the mechanism is the Euler identity $\sum_{L\ge1} p_L\, p_L^{\perp} = n \cdot \mathrm{id}$ on Λ_n together with Murnaghan–Nakayama (Theorem 3.1). This replaces a cited black box by a three-line proof.
- **Main theorem** (Theorem 7.2): for $\mu = (n)$ and every $n \ge 4$ there is an explicit certificate $c^{(n)}$ with $c^{(n)}M = 0$ and $c^{(n)}b = t(t + 1) \ne 0$, supported on the n hooks of size n together with the single row $(2, 2, 1^{n-4})$. Its $n = 4$ specialisation is the certificate printed in the earlier paper. By conjugation the same holds for $\mu = (1^n)$. This is the certificate family that gap (1) of that paper asked for.
- **Consequently the corollary that motivated the whole computation survives, and is upgraded from verified to proved**: since the Khanna–Loehr local identity is required at *every* μ, one failing family suffices, and $\mu = (n)$ fails for all $n \ge 4$. The Adin–Bauer/Khanna–Loehr inversion reciprocity does not deform along t — now for all n, not for $n \le 7$.
- Corrected classification (Theorem 5.1, verified $n \le 9$): the system is consistent over $\mathbb{Q}(t)$ exactly for $n \le 3$ (all μ), for $n = 4$, $\mu = (2, 2)$, and for $n = 5$, $\mu \in \{(3, 2), (2, 2, 1)\}$.
- For general μ I reduce the statement to a finite, t-free graph criterion on the n cells of μ (Theorem 8.2), derived from the abacus, and verified equivalent for all μ with $n \le 9$. Two combinatorial statements about that graph remain open; they are stated precisely in §9.

1 The system

Fix $n \geq 1$. For $\lambda \vdash n$ and $1 \leq L \leq n$ let $S(\lambda, L)$ be the set of partitions $\gamma \vdash n-L$ such that λ/γ is a rim hook (border strip) of size L, and let $\mathrm{ht}(\lambda/\gamma)$ be one less than the number of rows it occupies. Put $\mathrm{wt}_A(\lambda, \gamma) = t^{\mathrm{ht}(\lambda/\gamma)}$.

Fix $\mu \vdash n$, take $T(\mu, L) = S(\mu, L)$, and consider the linear system in the unknowns $\{\mathrm{wt}_B(\mu, \gamma)\}$ over $\mathbb{Q}(t)$, with no sign, positivity or combinatorial constraint:

$$\sum_{L=1}^{n} \sum_{\gamma \in S(\lambda,L) \cap S(\mu,L)} t^{\mathrm{ht}(\lambda/\gamma)} \, \mathrm{wt}_B(\mu, \gamma) \;=\; \delta_{\lambda\mu}, \qquad \forall\, \lambda \vdash n. \tag{1}$$

Since γ determines $L = n - |\gamma|$, the outer sum is bookkeeping: set

$$C(\mu) \;:=\; \{\gamma \;:\; \mu/\gamma \text{ is a rim hook}\} \;=\; \bigcup_{L=1}^{n} S(\mu, L),$$

let $M = M^{(\mu)}(t)$ be the matrix with rows indexed by $\lambda \vdash n$, columns by $\gamma \in C(\mu)$, and

$$M_{\lambda\gamma} \;=\; \begin{cases} t^{\mathrm{ht}(\lambda/\gamma)} & \lambda/\gamma \text{ a rim hook}, \\ 0 & \text{otherwise}, \end{cases}$$

and let $b = e_\mu$. Then (1) is $Mw = b$.

It is convenient to read the columns inside Λ_n, the degree-n part of the ring of symmetric functions with its Schur basis. Define the *L-rim-hook adding operator*

$$R_L(t)\, s_\gamma \;=\; \sum_{\lambda/\gamma \text{ rim hook of size } L} t^{\mathrm{ht}(\lambda/\gamma)}\, s_\lambda.$$

Then column γ of M is the coordinate vector of $R_{n-|\gamma|}(t)\, s_\gamma$, and

Observation 1.1. (1) is consistent over $\mathbb{Q}(t)$ if and only if $s_\mu \in \mathrm{span}_{\mathbb{Q}(t)}\{\, R_{n-|\gamma|}(t)\, s_\gamma \;:\; \gamma \in C(\mu)\,\}$.

2 The system is $p(n) \times n$

Proposition 2.1. *$|C(\mu)| = n$ for every $\mu \vdash n$. Hence (1) has $p(n)$ equations and exactly n unknowns.*

Proof. Fix $N > n + \ell(\mu)$ and let $B = B(\mu) = \{\mu_i + N - i \;:\; 1 \leq i \leq N\}$ be the beta-set (first-column hook lengths of μ padded to N beads). Removing a rim hook of size L from μ is the same as choosing a bead $b \in B$ with $b - L \geq 0$ and $b - L \notin B$ and replacing b by $b - L$; the resulting beta-set is $B(\gamma)$, and $\mathrm{ht}(\mu/\gamma) = |B \cap (b - L, b)|$. So $C(\mu)$ is in bijection with the set of pairs (b, b') with $b \in B$, $b' \notin B$, $0 \leq b' < b$.

Fix a bead $b = b_i = \mu_i + N - i$, i.e. a row i of μ. The classical first-column hook-length identity says that the set of holes below b_i, namely $\{0, 1, \ldots, b_i - 1\} \setminus B$, is exactly $\{\, b_i - h_{ij} \;:\; 1 \leq j \leq \mu_i \,\}$, where h_{ij} is the hook length of the cell (i, j): indeed $b_i - h_{ij}$ is strictly increasing in j, lies in $[0, b_i)$, and $b_i - h_{ij} \in B$ would force a cell of μ to the right of (i, j) in a row below i with the same hook, which is impossible. Counting both sides gives μ_i holes below b_i. Summing over i, the pairs (b, b') are in bijection with the cells of μ, of which there are n. $\square$

Remark 2.2. This corrects the explanation offered in `2026-09-10-Q129` for the nonzero entries of the "generic" column of its anchor scan ("for those μ the system happens to be square, $P(n)$ equations, $P(n)$ unknowns"). The system is 5×4 at $n = 4$ and 7×5 at $n = 5$; it is square precisely when $p(n) = n$, i.e. for $n \leq 3$.

3 The anchor $t = -1$ is the Euler identity

Theorem 3.1. *For every $n \geq 1$ and every $\mu \vdash n$, the vector $\mathrm{wt}_B(\mu, \gamma) = \frac{1}{n}(-1)^{\mathrm{ht}(\mu/\gamma)}$, $\gamma \in C(\mu)$, solves (1) at $t = -1$.*

Proof. Identify $\Lambda = \mathbb{Q}[p_1, p_2, \dots]$. Multiplication by p_L has Hall adjoint $p_L^{\perp} = L\,\partial/\partial p_L$, so on Λ_n

$$\sum_{L \geq 1} p_L\, p_L^{\perp} \;=\; \sum_{L \geq 1} L\, p_L \frac{\partial}{\partial p_L} \;=\; n \cdot \mathrm{id},$$

because $L\, p_L \partial_{p_L}$ acts on a monomial p_ν by $L\, m_L(\nu)$, and $\sum_L L\, m_L(\nu) = |\nu| = n$. By Murnaghan–Nakayama, $R_L(-1)$ is multiplication by p_L, and dually $p_L^{\perp} s_\mu = \sum_{\gamma \in S(\mu,L)} (-1)^{\mathrm{ht}(\mu/\gamma)} s_\gamma$. Hence

$$n\, s_\mu = \sum_{L=1}^{n} p_L\, p_L^{\perp} s_\mu = \sum_{L=1}^{n} \sum_{\gamma \in S(\mu,L)} (-1)^{\mathrm{ht}(\mu/\gamma)}\, R_L(-1)\, s_\gamma = \sum_{\gamma \in C(\mu)} (-1)^{\mathrm{ht}(\mu/\gamma)}\, R_{n-|\gamma|}(-1)\, s_\gamma.$$

By Observation 1.1 this is exactly (1) at $t = -1$, after dividing by n. ☐

Remark 3.2. This is the whole content of the $t = -1$ anchor: the Adin–Bauer/Khanna–Loehr inversion at $t = -1$ is the Euler degree operator seen through Murnaghan–Nakayama, and the inverse weights are just the MN signs divided by n. Any certificate of inconsistency over $\mathbb{Q}(t)$ must therefore have its obstruction value vanish at $t = -1$ — see Remark 7.5.

4 The theorem is false as stated

Theorem 4.1. *At $n = 4$, $\mu = (2, 2)$, the system (1) is consistent over $\mathbb{Q}(t)$.*

Proof. $C((2,2)) = \{(2,1), (2), (1,1), (1)\}$ with ht equal to $0, 0, 1, 1$ respectively; the matrix and right-hand side are

λ	$w_{(1)}$	$w_{(1,1)}$	$w_{(2)}$	$w_{(2,1)}$	rhs
(4)	1	0	1	0	0
$(3,1)$	0	1	0	1	0
$(2,2)$	t	t	1	1	1
$(2,1,1)$	0	0	t	1	0
$(1,1,1,1)$	t^2	t	0	0	0

Put $w = \big(-1,\ t,\ 1,\ -t\big)/(t-1)^2$. Then row (4) gives $(-1+1)/(t-1)^2 = 0$; row $(3,1)$ gives $(t-t)/(t-1)^2 = 0$; row $(2,1,1)$ gives $(t-t)/(t-1)^2 = 0$; row (1^4) gives $(-t^2 + t \cdot t)/(t-1)^2 = 0$; and row $(2,2)$ gives $(-t + t^2 + 1 - t)/(t-1)^2 = (t-1)^2/(t-1)^2 = 1$. ☐

Likewise at $n = 5$, $\mu = (3, 2)$ has the solution $\big(-1,\, t,\, 1,\, t^2,\, -t\big)/(2t^2 - 2t + 1)$ on the columns $(1), (1,1), (3), (2,2), (3,1)$, and $\mu = (2,2,1)$ has $\big(-\frac{1}{t(t^2-2t+2)}, \frac{1}{t(t^2-2t+2)}, \frac{t}{t^2-2t+2}, -\frac{t}{t^2-2t+2}, \frac{1}{t^2-2t+2}\big)$ on $(1), (2), (1,1,1), (2,1,1), (2,2)$. Both were substituted back symbolically.

Where the error is, and where it is not. The instrument used here was built from scratch: rim-hook removal was implemented twice, once by beta-numbers and once by brute-force enumeration over Young diagrams with an edge-connectivity and no-2×2 test, and the two agree on all 240 pairs (λ, L) with $|\lambda| \leq 7$. It reproduces the 5×4 matrix and the certificate printed in `2026-09-10-Q129` exactly, and it reproduces that paper's anchor-scan table entry for entry:

n	$t=-1$	$t=0$	$t=1$	generic
4	5/5	3/5	0/5	**1/5**
5	7/7	4/7	2/7	**2/7**
6	11/11	6/11	0/11	0/11
7	15/15	9/15	0/15	0/15

So the defect is localised in the statement of `thm:local`, not in that paper's computations: its own verification section already contains the counterexample count, and its prose explains it away with a squareness claim that Proposition 2.1 refutes. The $\mu = (4)$ certificate it displays is correct and is reproved below as the $n = 4$ member of a family.

5 Corrected classification

Theorem 5.1 (verified, $n \le 9$)**.** *For $\mu \vdash n$, (1) is consistent over $\mathbb{Q}(t)$ if and only if*

$$n \le 3 \quad (\textit{all } \mu), \qquad \textit{or} \qquad n = 4,\ \mu = (2,2), \qquad \textit{or} \qquad n = 5,\ \mu \in \{(3,2),\ (2,2,1)\}.$$

In particular it is inconsistent for every $\mu \vdash n$ with $n \ge 6$.

This is a finite computation for $n \le 9$ (exact row reduction of the $p(n) \times (n+1)$ augmented matrix over $\mathbb{Q}(t)$; 95 values of μ). For $n \le 3$ the system is square by Proposition 2.1 and the "if" direction is generic invertibility. The next two sections prove the "only if" direction for $\mu = (n)$ and $\mu = (1^n)$, all $n \ge 4$; the general case is reduced in §8.

6 A reduction, and a t-free first-order criterion

Lemma 6.1. *If the matrix $M^{(\mu)}$ with the row $\lambda = \mu$ deleted has full column rank n, then (1) is inconsistent over $\mathbb{Q}(t)$.*

Proof. The deleted rows are exactly the equations with right-hand side 0. If they force $w = 0$, the row $\lambda = \mu$ reads $0 = 1$. □

The converse also held in every case computed, $n \le 8$; I do not use it.

Deleting the row μ is a genuine simplification because the row μ is the only row all of whose entries are nonzero: by definition every $\gamma \in C(\mu)$ satisfies μ/γ a rim hook.

There is a second reduction, which strips t out of the problem altogether. Suppose $\operatorname{rank} M^{(\mu)}(-1) = n$ (verified for all μ with $n \le 8$). Then a fixed $n \times n$ minor is invertible at $t = -1$, the candidate solution $w(t)$ obtained from it by Cramer is regular at $t = -1$, and $w(-1)$ is the Euler solution of Theorem 3.1. The residual $r(t) = M(t)w(t) - b$ is regular at $t = -1$ with $r(-1) = 0$, and consistency over $\mathbb{Q}(t)$ is equivalent to $r \equiv 0$. Differentiating at $t = -1$:

Proposition 6.2. *Assume* $\operatorname{rank} M^{(\mu)}(-1) = n$*. Let* $H_L\, s_\gamma := \sum_{\lambda/\gamma \text{ rim hook of size } L} \operatorname{ht}(\lambda/\gamma)\,(-1)^{\operatorname{ht}(\lambda/\gamma)}\, s_\lambda$ *and put*

$$\Omega(\mu) \ := \ \sum_{L=1}^{n} H_L\, p_L^{\perp} s_\mu, \qquad W(\mu) \ := \ \operatorname{span}_{\mathbb{Q}}\{\, p_{n-|\gamma|} s_\gamma \ : \ \gamma \in C(\mu)\,\}.$$

If $\Omega(\mu) \notin W(\mu)$ then (1) is inconsistent over $\mathbb{Q}(t)$.

Everything in Proposition 6.2 is over $\mathbb{Q}$: the parameter t has been eliminated. Computationally the criterion is not merely sufficient but exact: $\Omega(\mu) \in W(\mu)$ if and only if (1) is consistent, for all 65 values of μ with $n \le 8$. The first-order term at the anchor decides the whole question.

7 Main theorem: the certificate family at $\mu = (n)$

Lemma 7.1. *Let $n \geq 2$ and $\mu = (n)$. Then $C((n)) = \{(k) : 0 \leq k \leq n-1\}$, all of height 0; write $w_k := \mathrm{wt}_B((n),(k))$. Let $\lambda \vdash n$.*

1. *If $\lambda = (n)$, the row of M is $(1,1,\ldots,1)$.*
2. *If $\lambda_3 \geq 2$, the row of M is zero.*
3. *Otherwise $\lambda = (a,b,1^m)$ with $a \geq b \geq 1$, $m \geq 0$, $a+b+m = n$, and the row has exactly two nonzero entries: t^{m+1} in column $k = b-1$ and t^m in column $k = a$.*

Proof. (1) Removing an L-rim hook from a single row gives $(n-L)$, of height 0, so $C((n)) = \{(k)\}$ and the row $\lambda = (n)$ has $(n)/(k)$ a horizontal strip in one row, height 0.

For the other parts fix k with $0 \leq k \leq n-1$ and ask when $\lambda/(k)$ is a rim hook. Necessarily $\lambda_1 \geq k$. The skew shape $\lambda/(k)$ contains all of rows $2,3,\ldots$ in full. If $\lambda_3 \geq 2$ then rows 3, 4 meet in a 2×2 block when $\lambda_4 \geq 2$, and in any case rows 2, 3 contain the 2×2 block on columns 1, 2 because $\lambda_2 \geq \lambda_3 \geq 2$; so no rim hook, which is (2).

So assume $\lambda_3 \leq 1$ and $\lambda_2 \geq 1$: $\lambda = (a,b,1^m)$ with $a = \lambda_1$, $b = \lambda_2 \geq 1$, $m = \ell(\lambda) - 2$. There are two cases.

$a > k$. Row 1 of the strip is the cells $(1,k+1),\ldots,(1,a)$; row 2 is $(2,1),\ldots,(2,b)$. Absence of a 2×2 block on rows 1, 2 forces $b \leq k+1$; connectivity of rows 1 and 2 forces $b \geq k+1$. Hence $b = k+1$, i.e. $k = b-1$. The strip occupies rows $1,\ldots,m+2$, so $\mathrm{ht} = m+1$.

$a = k$. Row 1 contributes nothing and the strip is rows $2,\ldots,m+2$: the cells $(2,1),\ldots,(2,b)$ together with the column $(3,1),\ldots,(m+2,1)$. This is connected and contains no 2×2 block. It occupies $m+1$ rows, so $\mathrm{ht} = m$, in column $k = a$.

Since $b - 1 < b \leq a$, the two columns are distinct, and no other k occurs. $\square$

Theorem 7.2. *Let $n \geq 4$ and $\mu = (n)$. Define $c = c^{(n)}$, indexed by $\lambda \vdash n$, by*

$$c_{(n)} = t(t+1), \qquad c_{(2,2,1^{n-4})} = \frac{1-(n-1)t}{t^{n-4}} \quad (\textit{added to any other value at that } \lambda),$$

$$c_{(a,1^{n-a})} = \frac{\kappa_a}{t^{n-a-1}} \quad (1 \leq a \leq n-1), \qquad \kappa_1 = (n-2)t^2 - 2t, \quad \kappa_2 = -t^2 + (n-2)t - 1, \quad \kappa_a = -t(t+1) \ (a \geq 3),$$

and $c_\lambda = 0$ for all other λ. Then

$$c\, M^{(n)} = 0 \qquad \textit{and} \qquad c\, b = t(t+1) \neq 0 \textit{ in } \mathbb{Q}(t).$$

Hence (1) is inconsistent over $\mathbb{Q}(t)$ for $\mu = (n)$ and every $n \geq 4$.

Proof. By Lemma 7.1 the equations of the homogeneous system, after dividing each by the common power of t in its row, are exactly

$$E_{a,b} \ : \quad t\, w_{b-1} + w_a = 0 \qquad \text{for all } a \geq b \geq 1 \text{ with } a+b \leq n,$$

together with $G : \sum_{k=0}^{n-1} w_k = 1$. Indeed the row $(a,b,1^m)$ is $t^{m+1} w_{b-1} + t^m w_a = 0$ with $m = n-a-b \geq 0$.

Take $b = 1$: $E_{a,1}$ reads $t\, w_0 + w_a = 0$ and is available for every $1 \leq a \leq n-1$. Take $a = b = 2$: $E_{2,2}$ reads $t\, w_1 + w_2 = 0$ and is available because $2+2 \leq n$. From $E_{1,1}$ and $E_{2,1}$ we get $w_1 = w_2 = -t\, w_0$; substituting into $E_{2,2}$,

$$0 = t\, w_1 + w_2 = -t^2 w_0 - t\, w_0 = -t(t+1)\, w_0,$$

so $w_0 = 0$ in $\mathbb{Q}(t)$, hence $w_a = 0$ for all a, contradicting G. This proves inconsistency; it remains to record the linear combination.

Writing the combination as $\alpha G + \sum_{a=1}^{n-1} \kappa_a E_{a,1} + d\, E_{2,2}$, the coefficient of w_0 is $\alpha + t \sum_a \kappa_a$, of w_1 is $\alpha + \kappa_1 + dt$, of w_2 is $\alpha + \kappa_2 + d$, and of w_a $(a \geq 3)$ is $\alpha + \kappa_a$. Setting all four to zero gives $\kappa_a = -\alpha$ for $a \geq 3$, $\kappa_1 = -\alpha - dt$, $\kappa_2 = -\alpha - d$ and then

$$\alpha + t\bigl(-(n-1)\alpha - d(t+1)\bigr) = 0 \quad\Longleftrightarrow\quad \alpha\bigl(1-(n-1)t\bigr) = d\,t(t+1).$$

Choosing $d = 1-(n-1)t$ and $\alpha = t(t+1)$ gives the stated κ's, and the right-hand sides combine to $\alpha \cdot 1 = t(t+1)$. Undoing the division by t^m turns $E_{a,1}$ into the row $(a, 1^{n-a})$ with $m = n-a-1$ and $E_{2,2}$ into the row $(2,2,1^{n-4})$ with $m = n-4$, which is the displayed c. □

Corollary 7.3. (1) *is inconsistent over* $\mathbb{Q}(t)$ *for* $\mu = (1^n)$ *and every* $n \geq 4$.

Proof. Conjugation $\lambda \mapsto \lambda'$ carries rim hooks to rim hooks and a rim hook of size L occupying r rows to one occupying $L - r + 1$ rows, so $\operatorname{ht}(\lambda'/\gamma') = L - 1 - \operatorname{ht}(\lambda/\gamma)$. Hence $M^{(\mu')}(t) = P\,M^{(\mu)}(1/t)\,D$ with P the permutation $\lambda \mapsto \lambda'$ of rows and $D = \operatorname{diag}\bigl(t^{\,n-|\gamma|-1}\bigr)$, while b is carried to b. Therefore $c'_{\lambda'} := c^{(n)}_\lambda\big|_{t\to 1/t}$ satisfies $c' M^{(1^n)}(t) = 0$ and $c'b = \frac{1}{t}\bigl(\frac{1}{t}+1\bigr) = \frac{1+t}{t^2} \neq 0$. □

Corollary 7.4 (the motivating statement, now proved for all n)**.** *For every $n \geq 4$ there is a $\mu \vdash n$ — namely $\mu = (n)$ — for which the Khanna–Loehr local identity with $T = S$ and free weights in $\mathbb{Q}(t)$ has no solution. Since the local identity is required at every μ, the Adin–Bauer/Khanna–Loehr inversion reciprocity does not deform along t: it holds at $t = -1$, where $R_e(-1)$ is multiplication by p_e and the statement is Theorem 3.1, and it fails for all $n \geq 4$.*

Remark 7.5. The obstruction value is $t(t+1)$, *independently of* n. The factor $(t+1)$ is not an independent sighting of the $(1+t)$ that appears in the sorting/commuting criterion: by Theorem 3.1 the system is consistent at $t = -1$ for every μ and every n, so *every* certificate whatsoever must have $c\,b$ vanish at $t = -1$. The factor is forced by a theorem already in hand and carries no information. That is the separator; the two $(1+t)$'s should be counted as one witness, not two.

The earlier paper's $n = 4$ certificate is $t \cdot c^{(4)}$, which is why its obstruction value reads $t^2(t+1)$ rather than $t(t+1)$; the extra root at $t = 0$ is an artefact of that normalisation, consistent with the remark in that paper that $t = 0$ is an accident of $\mu = (4)$.

8 General μ: reduction to a graph criterion

The mechanism of Theorem 7.2 is visible on the abacus and generalises to a criterion for all μ. Use the $\mathbb{Z}$-Maya convention: $B = B(\mu) \subset \mathbb{Z}$, beads at $b_i = \mu_i - i$, all positions far to the left occupied. By Proposition 2.1 the columns are the pairs (b, b') with $b \in B$ a bead and $b' \notin B$ a hole, $b' < b$; write $w(b, b')$ for the unknown.

A row $\lambda \neq \mu$ of M arises from a bead move down followed by a bead move up of the same size, so $\lambda = B \setminus \{b, c\} \cup \{b', e\}$ with $b, c \in B$, $b', e \notin B$ and $b + c = b' + e$. Sorting the four positions shows that such a λ generically admits exactly *two* decompositions of this kind, so it contributes a two-term relation. Computing the two heights gives:

Proposition 8.1 (reflection relations)**.** *Let $\mu \vdash n$ with Maya diagram B.*

(I) Let $c < b$ be beads and $b' < c$ a hole, and suppose $e := b + c - b'$ is a hole (it is automatically $> b$). Then M has a row with exactly the two entries in columns (b, b') and (c, b'), giving

$$w(c, b') \;=\; -\,t^{\,|B\cap(c,b)|}\, w(b, b').$$

(II) *Let b be a bead and $b' < e < b$ holes, and suppose $c := b' + e - b$ is a bead (it is automatically $< b'$). Then M has a row with exactly the two entries in columns (b, b') and (b, e), giving*

$$w(b,e) \;=\; -\,t^{\,1+|B\cap(b',e)|}\,w(b,b').$$

These were checked against the matrix: for every μ with $n \le 8$, the relations of Proposition 8.1 span exactly the same space as the set of *all* rows of M having exactly two nonzero entries.

Let $G(\mu)$ be the graph on the n columns (equivalently, by Proposition 2.1, on the n cells of μ) whose edges are the relations (I) and (II), each edge labelled by its ratio $-t^h \in \mathbb{Q}(t)^\times$.

Theorem 8.2. *If every connected component of $G(\mu)$ contains a cycle whose product of edge labels (traversed with signs) is $\neq 1$ in $\mathbb{Q}(t)$, then $w = 0$ is forced by the rows $\lambda \neq \mu$, and (1) is inconsistent by Lemma 6.1.*

Proof. On a component, the relations determine each w from any one of them, up to a unit $\pm t^k$. A cycle with label product $\neq 1$ forces that w, hence the whole component, to vanish. ☐

Proposition 8.3 ($\mu = (n)$, in this language)**.** *For $\mu = (n)$ the Maya diagram has one bead at n above a sea of beads at all positions < 0, and holes at $0, \ldots, n-1$; the columns are (n, k), $k = 0, \ldots, n-1$. Only relations of type (II) occur, and the condition "$c = k + j - n$ is a bead" reads $k + j \le n - 1$, with $|B \cap (k, j)| = 0$. So the edges are $\{k, j\}$ with $k + j \le n - 1$, all with label $-t$. The vertex 0 is adjacent to all others, and for $n \ge 4$ the triangle $\{0, 1, 2\}$ is present with label product $(-t)^3 \neq 1$ around it — equivalently $-t \neq t^2$, i.e. $t(t+1) \neq 0$. This is Theorem 7.2 again.*

Verification. For every μ with $n \le 9$ (95 partitions): $G(\mu)$ is connected; and the criterion of Theorem 8.2 holds exactly when (1) is inconsistent — it fails precisely at $\mu = (2,2)$, $(3,2)$, $(2,2,1)$ and for $n \le 3$.

9 Gaps, precisely stated

1. **Theorem 5.1 is verified, not proved, for $\mu \notin \{(n), (1^n)\}$.** The proved statements are Theorem 7.2 and Corollary 7.3: all $n \ge 4$ at the two extreme μ. Everything else is a finite computation, $n \le 9$.

2. **Two combinatorial statements about $G(\mu)$ remain open**, and together they would finish Theorem 5.1:

   (G1) $G(\mu)$ is connected for every μ. (Verified $n \le 9$.)

   (G2) For $n \ge 6$, $G(\mu)$ contains a cycle of nontrivial label product. (Verified $n \le 9$; false exactly at $(2,2), (3,2), (2,2,1)$ and $n \le 3$.)

   For $\mu = (n)$ both are elementary, as above. For general μ I do not have them. I note that the type-(I) exponent is simply $i' - i - 1$ when the two beads are the rows $i < i'$ of μ, and that a type-(II) edge between two cells of row i with hooks $d_1 > d_2$ exists as soon as $d_1 + d_2 > h_{i1}$ — so the corner cell $(1,1)$ is joined to every other cell of row 1. That gives a hub but not, by itself, a triangle: for $\mu = (3,1,1,1,1)$ the row-1 hooks are $7, 2, 1$ and no second edge inside row 1 exists.

3. **Two routes were tried and refuted.** (a) The brief's suggestion that the obstruction lives on the hook rows: for μ not a hook, the n hook rows of $M^{(\mu)}$ have rank 3, not n — badly deficient, and not repairable by adding a bounded number of rows. (b) The family $\{\lambda : \lambda_3 \leq 1\}$, which is everything for $\mu = (n)$: it has full column rank for many μ but fails for e.g. $(3,3)$, $(4,3)$, $(4,4)$, $(2,2,2,2)$.
4. **Proposition 6.2 carries a hypothesis I verified but did not prove**, namely rank $M^{(\mu)}(-1) = n$, i.e. linear independence of $\{p_{n-|\gamma|}s_\gamma : \gamma \in C(\mu)\}$ in Λ_n (checked for all μ, $n \leq 8$). It is not used in §7.
5. **Gap (2) of the earlier paper is untouched.** Khanna–Loehr allow $T(\mu, L)$ to be any subset of $R(n-L)$. Nothing here bears on that; Corollary 7.4 refutes only the symmetric choice, for all n rather than for $n \leq 7$.
6. **No novelty claim is made** for $R_e(t)$ or for any identity about it, in accordance with the standing block.

10 Verification

All computations are exact, in `sympy` over $\mathbb{Q}(t)$; no floating point. Scripts are in `scratch/q140/`.

- *Instrument, two independent implementations.* Rim-hook removal by beta-numbers and by brute-force enumeration over Young diagrams (edge-connectivity plus no-2×2, height read off the row set) agree on all 240 pairs (λ, L) with $|\lambda| \leq 7$, as sets and on every height.
- *Calibration before use.* The instrument reproduces the 5×4 matrix printed in `2026-09-10-Q129` entry for entry, its certificate ($cM = 0$, $cb = t^2(t+1)$), and its four-row anchor-scan table entry for entry.
- *Proposition 2.1:* $|C(\mu)| = n$ for all μ, $n \leq 9$.
- *Theorem 3.1:* the vector $(-1)^{\mathrm{ht}(\mu/\gamma)}/n$ substituted into the matrix at $t = -1$ gives b exactly, for all μ, $n \leq 8$.
- *Theorem 4.1:* solutions substituted back symbolically, $Mw - b = 0$, for all three exceptional μ.
- *Theorem 5.1:* exact row reduction of $[M \,|\, b]$ over $\mathbb{Q}(t)$ for all 95 partitions with $n \leq 9$.
- *Lemma 7.1:* predicted row compared entry by entry against the abacus-built matrix for $2 \leq n \leq 11$; exact.
- *Theorem 7.2:* $c^{(n)}M = 0$ and $c^{(n)}b = t(t+1)$ verified symbolically for $4 \leq n \leq 11$; at $n = 4$, $t\,c^{(4)}$ equals the certificate printed in `2026-09-10-Q129` on the nose.
- *Lemma 6.1:* rank of M with row μ deleted equals n exactly when (1) is inconsistent, for all μ, $n \leq 8$ (65 cases; no exceptions).
- *Proposition 6.2:* $\Omega(\mu) \in W(\mu)$ iff consistent, for all μ, $n \leq 8$; and rank $M(-1) = n$ throughout.
- *Proposition 8.1 and Theorem 8.2:* derived relations span the same space as the empirical two-term rows for all μ, $n \leq 8$; the graph criterion agrees with consistency for all μ, $n \leq 9$, and $G(\mu)$ is connected in all 95 cases.